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Assignment 1

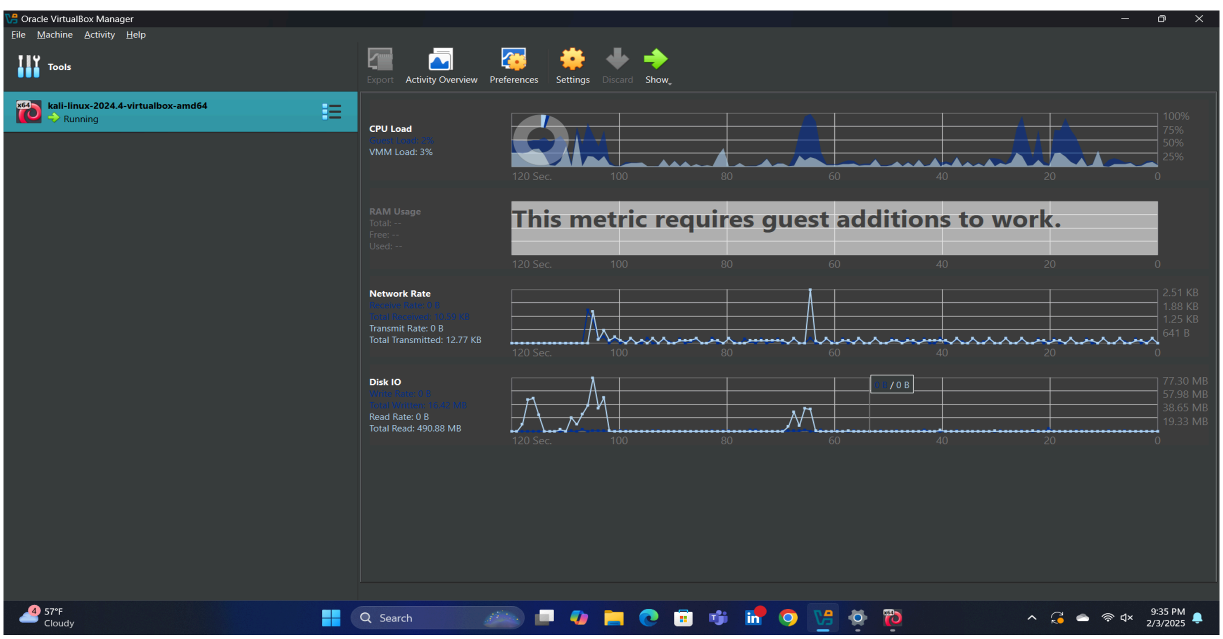
Abstract
Conducting a thorough analysis of network traffic to identify any information exposed during transit. In this project we generate and transmit sample traffic while closely monitoring the packets for content. The results indicate that all transmitted information remains unencrypted and can be easily intercepted and read, showing how unencrypted data can be vulnerable.

Introduction
Using kali linux as an operating system in a virtual box virtual machine; Using Netcat to create a network traffic for sending unencrypted data like text, and using wireshark to intercept and capture the network packets and also revealing the un-encrypted data.

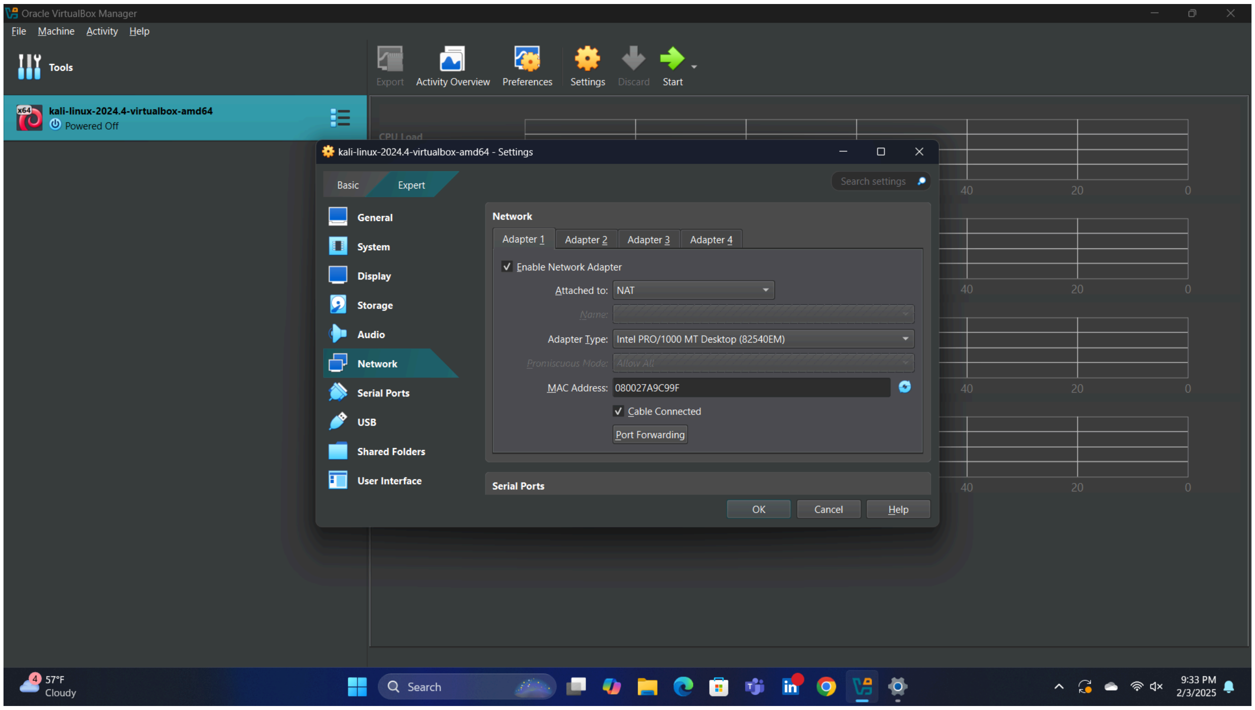
Netcat listener commands used:
nc -l 127.0.0.1 -p 31337
Netcat sender commands:
nc 127.0.0.1 -p 31337

WireShark runs as root while capturing on the ‘any’ adapter.

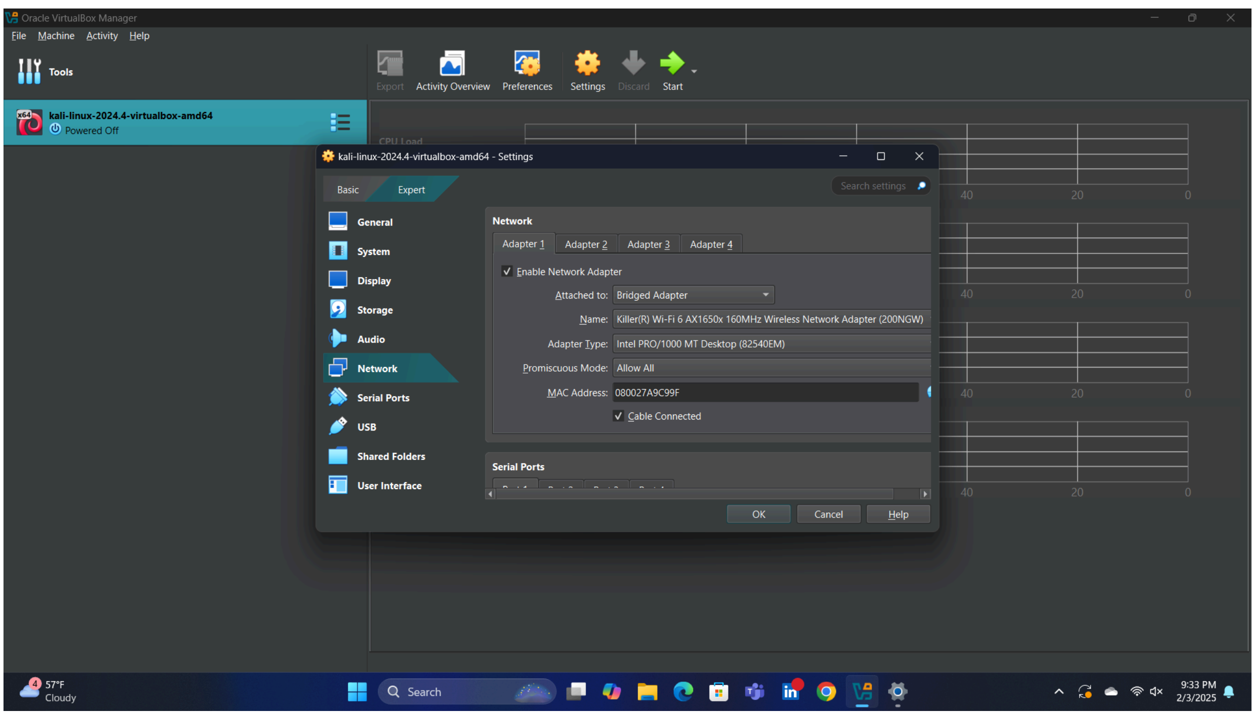
Summary of Results:
Downloading the kali linux virtual machine version and Using a Virtual box as a virtual machine in our windows system to run kali linux.



And before running kali Linux in VM, Change the network settings to bridged Adapter from Nat. Changing the network settings in a Kali Linux virtual machine allows the VM to act as if it is directly connected to the same network as the host machine.

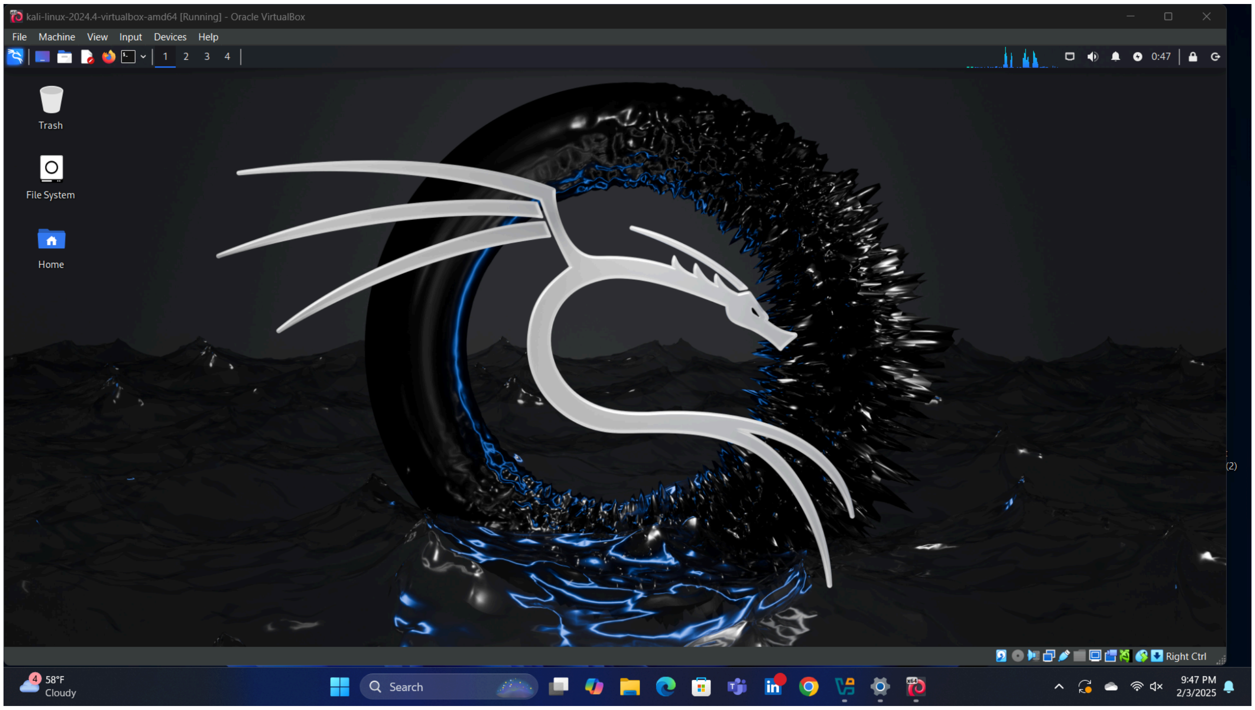


The Above picture shows that the virtual machine is connected to the NAT network.

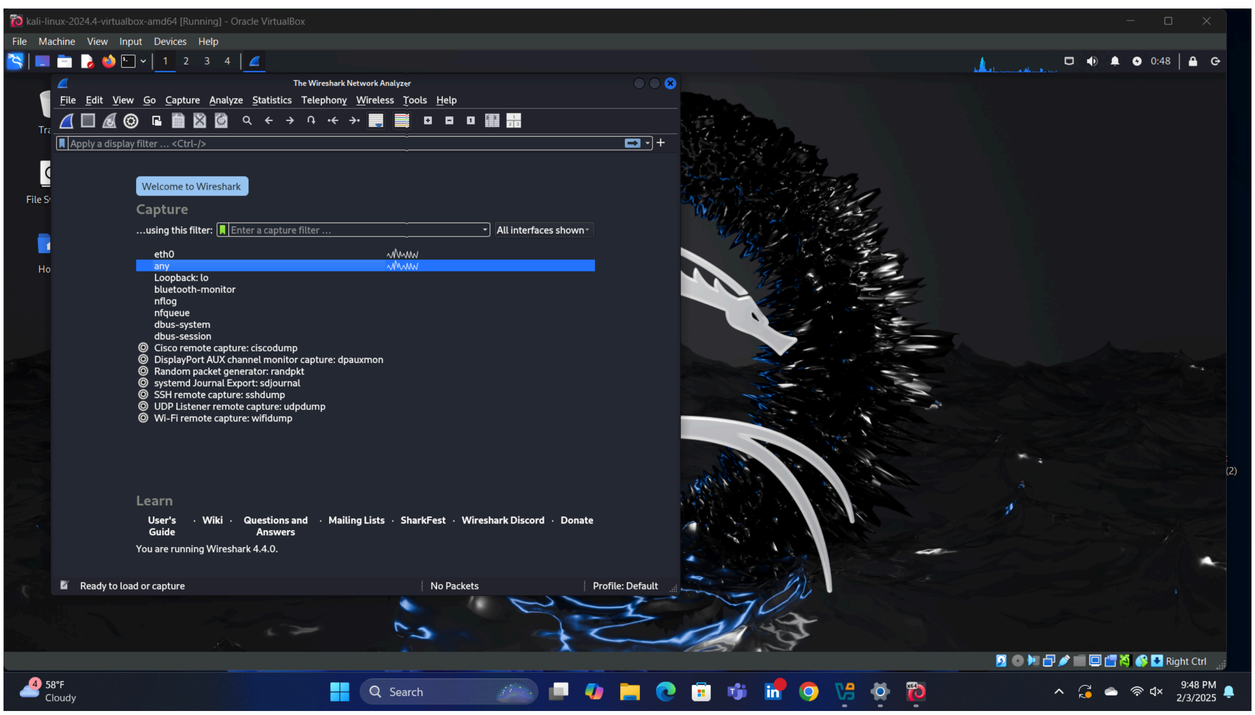


This Picture shows that the VM is connected to a Bridged adapter.
Now once the VM starts running kali linux, go to the top left corner and click the search box and search for Wire shark- A network intercepting tool. You can also find Wireshark inside the

Sniffing and spoofing tab. Open the Wireshark and select any as the filter to capture data packets.

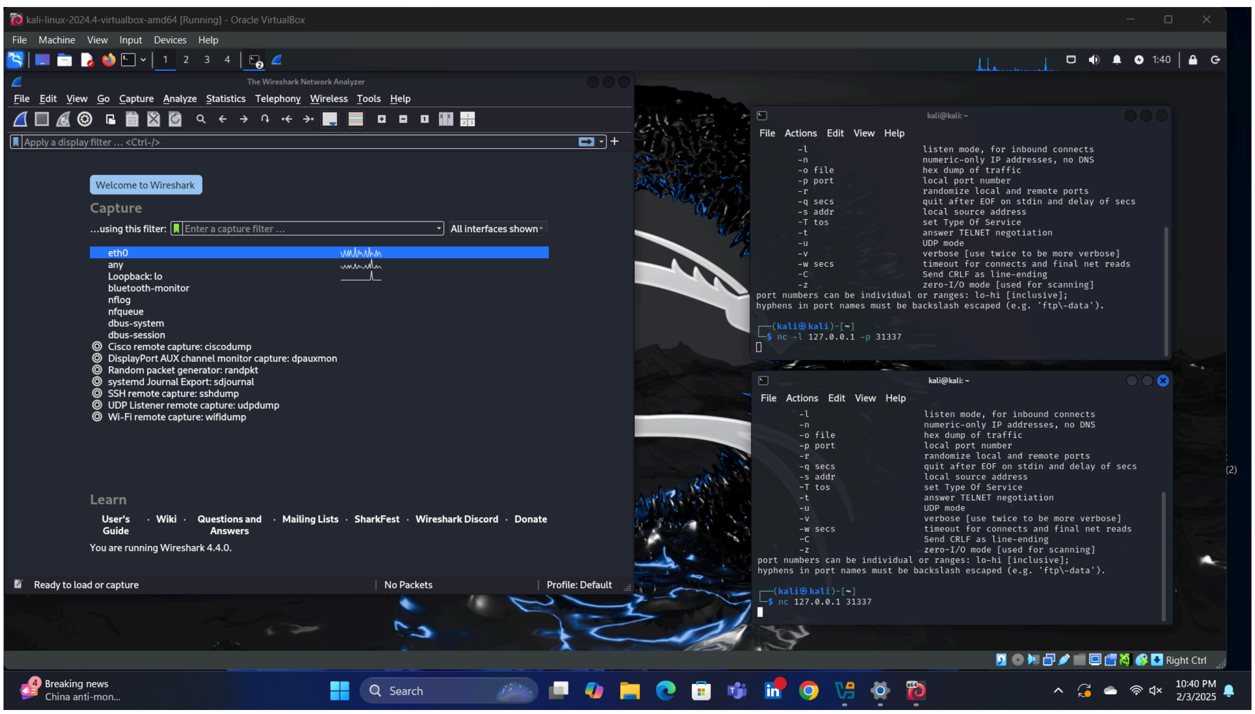


The Above picture is showing the Kali Linux operating system running on a VM.



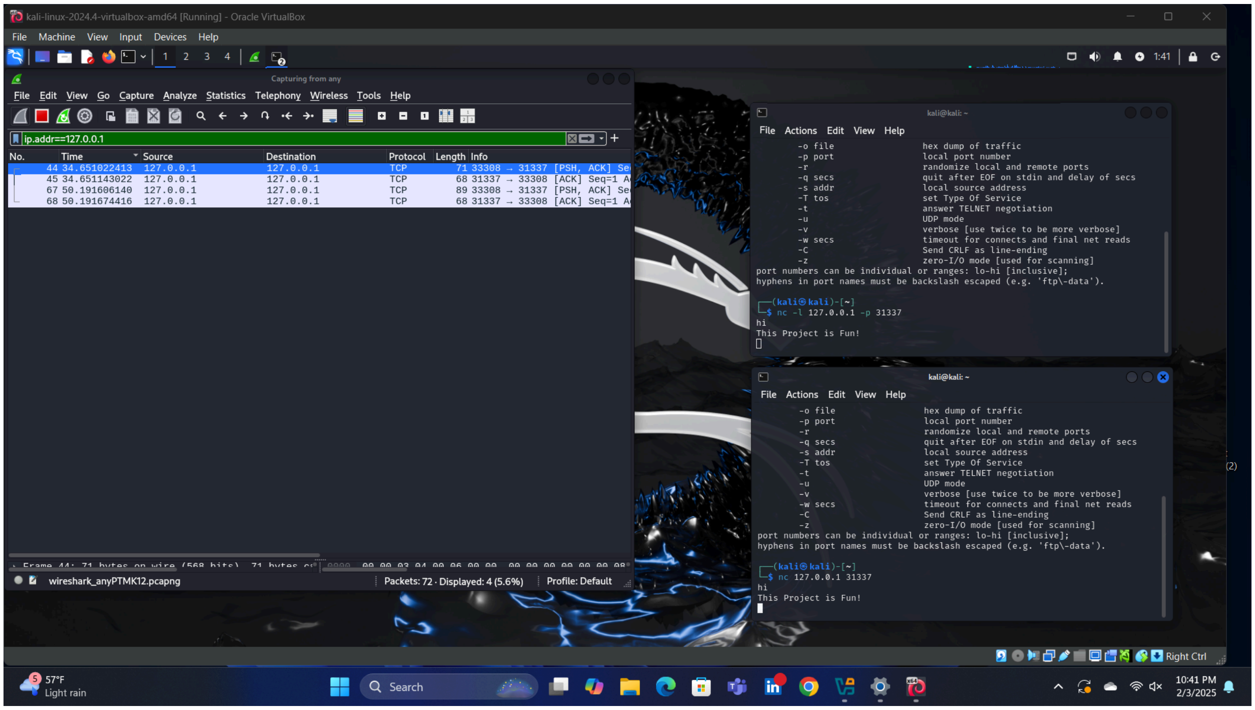
This picture shows the Wireshark application is open and the filter any is being selected for capturing the data packets.

Open Bash (Bourne Again Shell) which is a command-line interpreter in Kali linux. And type in “nc -h” in the command line, it will automatically open the Netcat(A tool used to manage networks, monitor traffic flow, and create simple unencrypted chat servers). You can also open that by going to search. After opening Netcat, Create a Netcat listener by typing the command line: nc -l 127.0.0.1 -p 31337. And then, open another Netcat interface which will be used as a sender, by typing Netcat sender command : nc 127.0.0.1 -p 31337. Here in this a network tunnel is created to the same ip network because 127.0.0.1, also known as "localhost", is the IP address of a local computer. It's a loopback address that allows a computer to communicate with itself; we are creating an un-encrypted network traffic inside the kali linux OS.

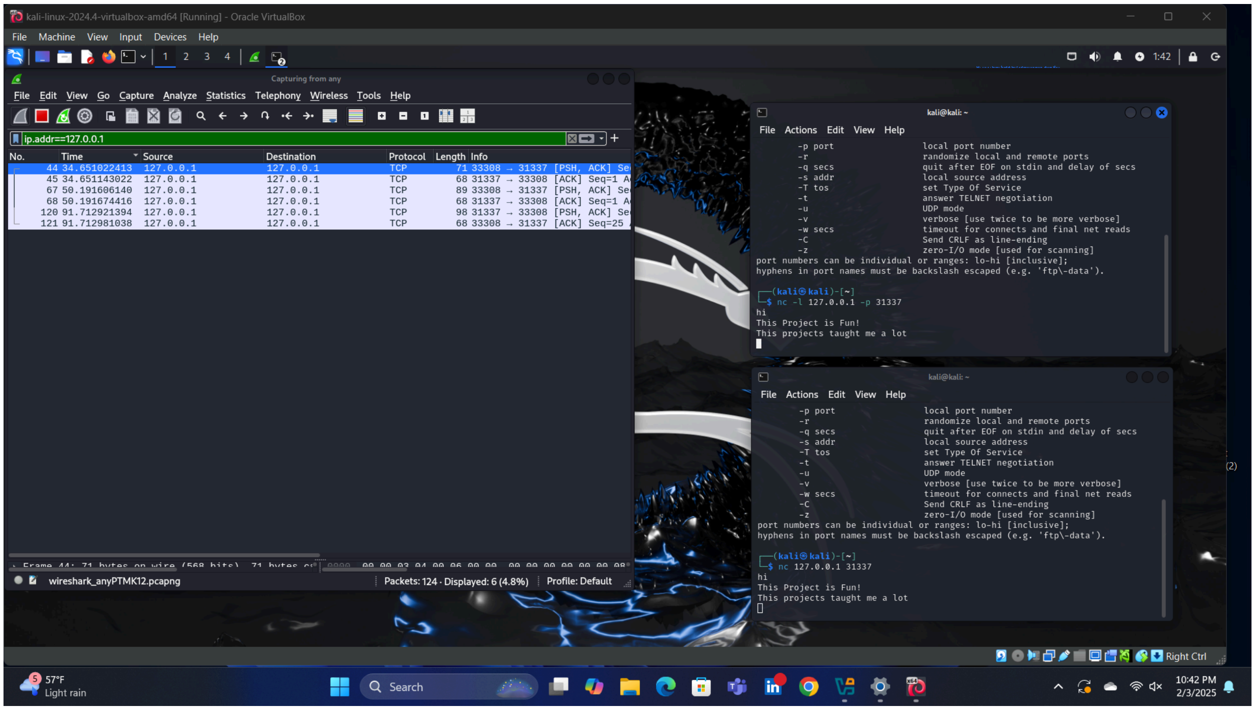


The above picture is showing the two Netcat terminals are open on the right, the top one being the listener and the bottom as sender. And towards the left side the wireshark is opened and ready for capturing network traffic.

Once the wireshark is opened and selected to “any” in the filter. It will start intercepting the Network and will be capturing all the network packets being performed on the network. This will collect both unencrypted and encrypted data packets. Now that the wireshark is running, go to the Netcat sender terminal and type anything like “hi, This project is fun! This project taught me a lot”. After typing the phrases in the sender terminal, you could see that these phrases are also appearing on the listener terminal. Hence this shows that the network traffic is being generated among the two terminals and its unencrypted. As you are typing the phrases in Netcat you could see that the Netshark is collecting all the packets of information. Once all the packets are collected, filter them out in the netshark by typing command “ ip.addr==127.0.0.1” this is the Loopback ip used between two Netcat terminals for communication. Once we filter the network packets of the ip address we could see the TCP network packets. By clicking each packet we could see the conversation between the two terminals.

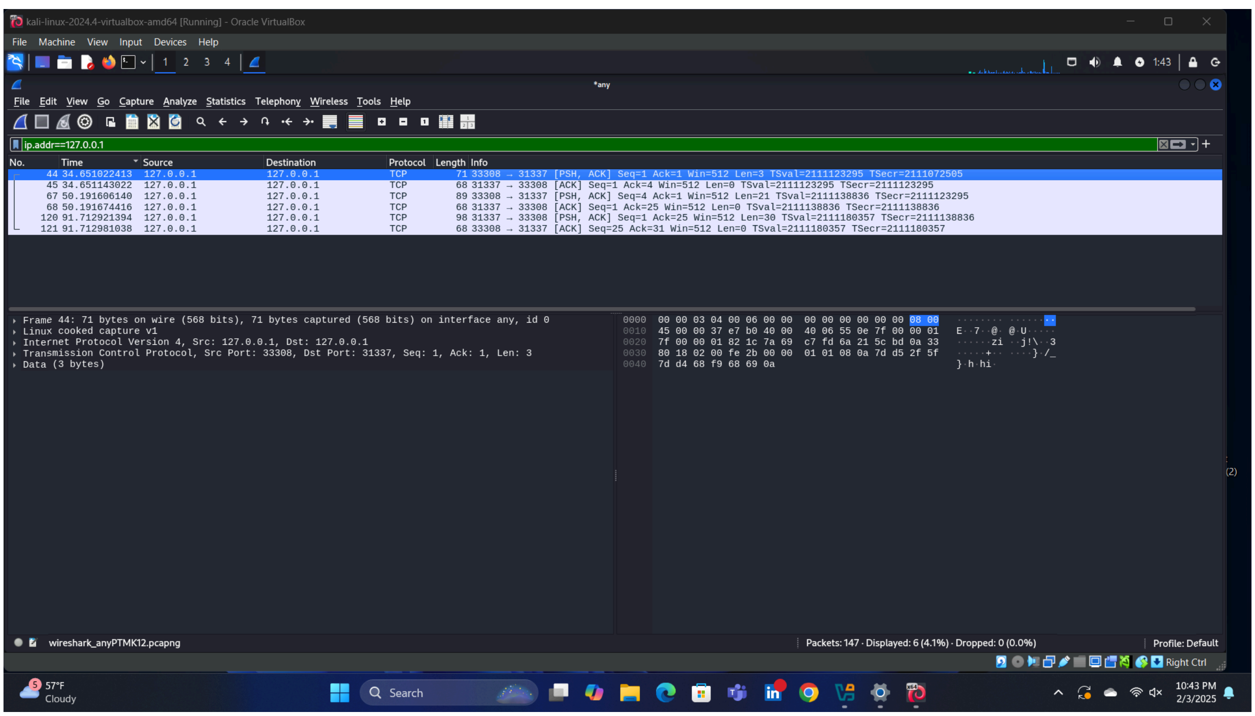


This picture shows the Wireshark application is capturing the data packets and the filter is set for a loopback ip, which is being used by Netcat. And also there is conversation being performed in the Netcat Terminals.

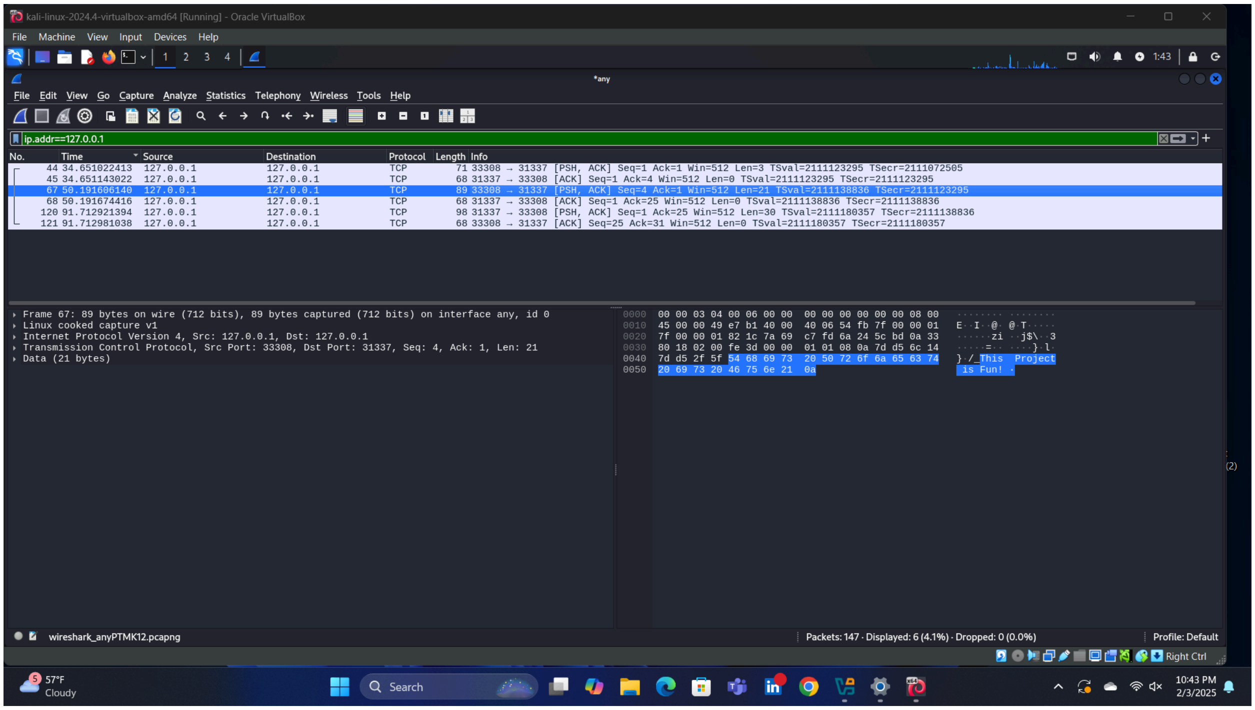


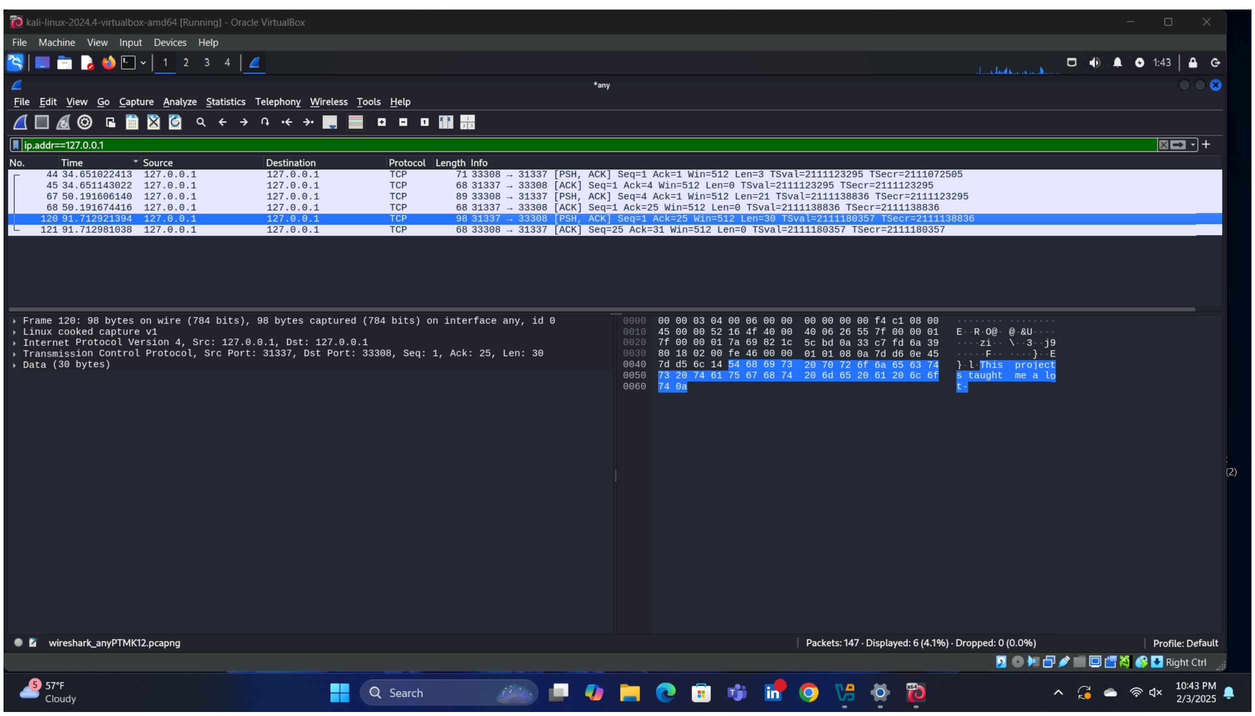
You could see that the network packets are growing as the conversation is more.

After pressing stop in Wireshark, go through the packets and check one by one; you could see the date and time, source and destination and size of data. Once you look at the bottom there is a button named data, once you click that you can clearly see what unencrypted data was transferred between two terminals.



The above picture is showing the first data packet conversation “hi” in the data section.





The above picture shows the next two conversations being performed by the Netcat terminals.

Conclusion:

Through this project, we got a clear understanding of how unencrypted data can be easily intercepted and read using Wireshark. By setting up a simple communication between two Netcat terminals in Kali Linux, and through Wireshark, we were able to capture and analyze network packets, revealing the transmitted text. Wireshark is a network analysis tool that allows users to capture and inspect network traffic in real-time. It works by putting a network interface into promiscuous mode, enabling it to capture all packets on the targeted network. Wireshark is mostly used to understand how data is transferred between devices or across the network. Moreover it helps to troubleshoot or to monitor networks for better performance. To intercept network communications, Wireshark listens to traffic on a selected network interface, such as Wi-Fi or Ethernet, and records all packets transmitted on the network. It allows users to filter the data packets by protocols, IP addresses, or specific types of data. This filtering helps to narrow down the data we are looking for like a specific ip address as used in the project. In the project I discovered that tools like wireshark can display detailed information about each captured packet, including the source and destination IP addresses, the protocol, and any data being sent on the network. If the data is unencrypted, Wireshark will display it, potentially exposing sensitive information. This makes it a powerful tool for detecting vulnerabilities, such as unsecured communications or weak encryption. This clearly demonstrates how vulnerable unencrypted data is and how easily an attacker could access sensitive information. This highlights the importance of encryption in network security. Without it, any data sent over a network can be intercepted, putting personal and organizational information at risk. Overall, this project was an eye-opening experience, reinforcing the need for strong encrypting practices. It showed how crucial it is to be aware of network vulnerabilities and take proactive steps to secure communications.